Progress Check

Identify a rationalising factor for each one of the following surds and verify the same in each case:

(i) $\sqrt{18}$ (ii) $5\sqrt{12}$ (iii) $\sqrt[3]{49}$ (iv) $\frac{1}{\sqrt{8}}$

2.7.1 Conjugate Surds

Can you guess a rationalising factor for $3+\sqrt{2}$? This surd has one rational part and one radical part. In such cases, the rationalising factor has an interesting form.

A rationalising factor for $3+\sqrt{2}$ is $3-\sqrt{2}$. You can very easily check this.

$$\begin{aligned}\left(3+\sqrt{2}\right)\left(3-\sqrt{2}\right) &= 3^2-\left(\sqrt{2}\right)^2\\ &= 9-2\\ &= 7, \text{ a rational.}\end{aligned}$$

What is the rationalising factor for $a+\sqrt{b}$ where a and b are rational numbers? Is it $a-\sqrt{b}$? Check it. What could be the rationalising factor for $\sqrt{a}+\sqrt{b}$ where a and b are rational numbers? Is it $\sqrt{a}-\sqrt{b}$? Or, is it $-\sqrt{a}+\sqrt{b}$? Investigate.

Surds like $a+\sqrt{b}$ and $a-\sqrt{b}$ are called conjugate surds. What is the conjugate of $\sqrt{b}+a$? It is $-\sqrt{b}+a$. You would have perhaps noted by now that a conjugate is usually obtained by changing the sign in front of the surd!

Example 2.27 Rationalise the denominator of (i) $\frac{7}{\sqrt{14}}$ (ii) $\frac{5+\sqrt{3}}{5-\sqrt{3}}$

Solution

(i) Multiply both numerator and denominator by the rationalising factor $\sqrt{14}$.

$$\frac{7}{\sqrt{14}}=\frac{7}{\sqrt{14}}\times\frac{\sqrt{14}}{\sqrt{14}}=\frac{7\sqrt{14}}{14}=\frac{\sqrt{14}}{2}$$

(ii) $$\begin{aligned}\frac{5+\sqrt{3}}{5-\sqrt{3}} &= \frac{\left(5+\sqrt{3}\right)}{\left(5-\sqrt{3}\right)}\times\frac{\left(5+\sqrt{3}\right)}{\left(5+\sqrt{3}\right)}=\frac{\left(5+\sqrt{3}\right)^2}{5^2-\left(\sqrt{3}\right)^2}\\ &= \frac{5^2+\left(\sqrt{3}\right)^2+2\times5\times\sqrt{3}}{25-3}\\ &= \frac{25+3+10\sqrt{3}}{22}=\frac{28+10\sqrt{3}}{22}=\frac{2\times[14+5\sqrt{3}]}{22}\\ &= \frac{14+5\sqrt{3}}{11}\end{aligned}$$

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